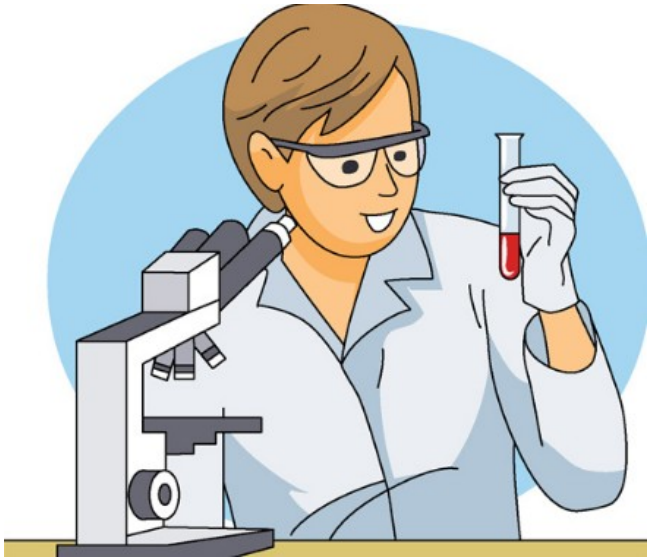
Science Inquiry



Intro to Scientific Method

Types of Investigation

What is science?

- Two aspects:
  - A process of inquiry
    - a way of finding out about things.
    - Scientists observe the world and come up with ideas:
      - About connections between things
      - About why things happen
    - Scientists use a systematic process “the scientific method” to explore connections and prove/disprove them.
  - A body of knowledge
    - Gained by systematic observation and testing of ideas.

In human biology and medical research…

- Inquiry, for example:
  - Formulating and testing drugs
  - Studying human health over time
  - Gathering data on human systems

- Leads to medical advances and more accurate understandings – contributes to a body of knowledge

Types of Investigation

- **Observations**
  - Using senses and measuring equipment
  - Gathering, collating and analysing data gathered
  - Looking for patterns and relationships within the data.

  Examples:
  - Observation of animal behaviour
  - Discovery of *Helicobacter pylori*



- **Surveys**
  - Collection of data from a large number of people.
  - Questionnaire or interview.
  - Can reveal patterns of behaviour, provide ideas about connections (eg lifestyle and heart disease).
  - Often used in conjunction with other data collection methods.



- **Trial and Error**
  - Systematic
  - Testing things in turn
  - Can be prolonged/time consuming
  - Keeping very accurate records

Example: testing different chemical compounds from plants, to see whether they have an antibiotic effect on bacterial colonies



- **Case Studies**
  - In depth investigation and report of a specific situation.
  - Can be used to track the progress of a disease in an individual, to learn more about disease in general, or to confirm other data.
  - Have limitations – can be too narrow in view, lack validity.

- **Longitudinal Studies**
  - Conducted over a long period
  - Gathers data repeatedly
  - Picture of change over time
  - Can establish links between lifestyle and health

    - Eg: Longitudinal studies of women's health.
    - Longitudinal monitoring of smoking habits and disease.

- **Controlled Experiments**
  - Designed to investigate relationships between factors.
  - Variables carefully controlled.
  - Validity carefully monitored.
  - Tests a hypothesis
  - Data undergoes statistical analysis to determine probability that hypothesis is correct.

  Eg: testing whether a particular drug is effective.